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Executive Summary
Multicellular spheroids have a remarkable capacity to mimic solid tumors to a greater extent than cell monolayers. Spatial organization within spheroids is seminal to their in vivo-like behavior and is instrumental in the acquisition of drug resistance and expression of antigens that can serve as biomarkers for carcinogenesis. Our research provides a comprehensive analysis of the spatial profile and assembly of multicellular spheroids, using prostate cancer cells as a model system. Spheroid differentiation and viability will be determined as a function of radial position. This analysis will be performed on spheroids produced using conventional methods and in a rotating-wall vessel (RWV). The knowledge gained from this research will ultimately improve the fidelity of multicellular spheroids in the extent to which they resemble micrometastases and avascular regions of larger tumors. In future work, we will use the spheroids to develop biomarkers for carcinogenesis and countermeasure to against the harmful effects of radiation.

Task Description:

Project Objective and Aims

The objective of the proposed research is to improve our understanding of how culture conditions affect the spatial profile and assembly of multicellular cancer spheroids, using prostate cancer cells as a model system. During the grant period, we fulfilled the following specific aims:

1) Developed mathematical models of spheroid formation with population balances and Monte Carlo technique.

2) Elucidated the biological and physical mechanisms underlying spheroid formation of prostate cancer cells with the mathematical models.

3) Assessed the ability of the mathematical models to predict spheroid self-assembly of prostate cancer cells.

4) Characterized the spatial composition of prostate cancer spheroids with respected to viability, differentiation and multidrug resistance.

NASA Critical Path Roadmap

The research described in this report is categorized in the NASA Critical Path Roadmap under the discipline of Radiation Health and addresses the risk of carcinogenesis in astronauts during the exploration of space from exposure to harmful radiation or from the synergistic effects of radiation with other space flight factors including stress. The relevant enabling question for this project is 31d: How can effective biomarkers of carcinogenic risk from space radiation be developed and validated?

The overall goal of this four-year project is to produce in vitro cultures of multicellular spheroids that mimic solid tumors by gaining an understanding of tissue assembly and spheroid composition. Through this understanding, we will identify culture conditions favorable to spheroid production. In future work, we will use the spheroids to develop biomarkers for carcinogenesis and countermeasure to against the harmful effects of radiation.

Background

For NASA, spheroids have been the primary form of tissue reconstructed in RWV's. For the larger tissue engineering community, spheroids have a variety of applications from liver regeneration to stem-cell differentiation. When spheroids are prepared from neoplastic cells as in the proposed research, they mimic micrometastases and avascular regions of larger tumors. Spheroids have proven beneficial as a research tool to investigate cancer progression, identify biomarkers of carcinogenesis, and in the testing of new therapies and delivery strategies to combat this deadly disease.

Numerous studies have demonstrated that the RWV produces multicellular spheroids that are more differentiated than cultures of the same cell line grown as a monolayer in terms of spheroid morphology and expression of proteins that are indicative of differentiated function. This is attributed to gentle mixing in the RWV that promotes mass transfer of nutrients and wastes while minimizing cellular damage from hydrodynamic forces; three-dimensional (3D) organization of cells that produces an environment in which cell shape, cell-cell and cell-matrix interactions, and interstitial fluid are akin to those in native tissue; and, for co-cultures, co-location of dissimilar cell types, mimicking the diversity found in vivo.

Understanding the mechanisms of spheroid assembly will ultimately improve the fidelity of this reconstructed tissue. The kinetic models developed in this work will be useful in producing spheroids within a defined size range that have greater viability. For, neoplastic cells, model results can also aid in the analysis of cellular adhesion and the effectiveness of new therapies to inhibit tumor shedding.

A critical component in the resemblance of spheroids to solid tumors is the spatial profiles within spheroids as a result of mass transport limitations and the 3D environment within the tissue. For example, the region of quiescent cells that lies beneath proliferative cells on the spheroid surface is often instrumental in the failure of tumor regression therapies. These cells have both limited accessibility and intrinsic resistance to toxic agents. Despite the importance of the spatial profile of spheroids to their function, there has been very little research done in this area with RWV's. In fact, ours will be by far the most comprehensive study of its kind for RWV spheroids.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Prostate cancer is one of the most common forms of cancer in men and a leading cause of death in the United States. While surgery and radiation will successfully treat many patients with localized disease, there is still no cure for patients with advanced, metastatic prostate cancer. The proposed research will aid in the development of new cancer therapies and biomarkers for detect prostate cancer by producing tissue models that more closely approximate micrometastases than is currently possible with conventional cell-culture techniques.

FINAL REPORT
The major findings of our study and their significance are summarized below. The data are organized chronologically by the peer-reviewed article in which they were published. (The symbol * designates corresponding author.)

RM Enmon, KC O'Connor*, H Song, DJ Lacks and DK Schwartz. Aggregation kinetics of well and poorly differentiated human prostate cancer cells. Biotechnol Bioeng, 80: 580-8 (2002).

Aggregation of attachment-dependent animal cells represents a series of motility, collision, and adhesion events applicable to such diverse fields as tissue engineering, bioseparations, and drug testing. Aggregation of human prostate cancer cells in liquid-overlay culture was modeled using Smoluchowski's collision theory. Using well (LNCaP) and poorly differentiated (DU 145 and PC 3) cell lines, the biological relevance of the model was assessed by comparing aggregation rates with diffusive and adhesive properties. Diffusion coefficients ranged from 5 to 90 microm(2)/min for single LNCaP and PC 3 cells, respectively. Similar diffusivities were predicted by the persistent random walk model and Einstein relation, indicating random motion. LNCaP cells were the most adhesive in our study with reduced cell shedding, 100% adhesion probability, and enhanced expression of E-cadherin. There was an increase in DU 145 cells staining positive for E-cadherin from nearly 20% of single cells to uniform staining across the surface of all aggregates;

Task Progress:

under 30% of PC 3 aggregates stained positive. Aggregation rates were more consistent with adhesive properties than with motilities, suggesting that aggregation in our study was reaction-controlled. Relative to other assays employed here, aggregation rates were more sensitive to phenotypic differences in cell lines and described size-dependent changes in aggregation at a finer resolution. In particular, model results suggest similar aggregation rates for two-dimensional DU 145 and PC 3 aggregates and upwards of 4-fold higher rates for larger three-dimensional DU 145 spheroids, consistent with expression of E-cadherin. The kinetic model has application to spheroid production, to cell flocculation and as an adhesion assay.

H Song, KC O'Connor*, O David, CL Giordano, H Pappas-LeBeau and S Clejan. Immunohisto-chemical analysis of differentiation in mixed and static prostate cancer spheroids. J Cell Mol Med, 7: 180-6 (2003).

Neoplastic multicellular spheroids are in vitro models of solid tumors employed in drug testing and basic research. This study compares differentiation in static and mixed prostate cancer spheroids. Staining intensity of prostate specific antigen (PSA) was down-regulated upon mixing, from 0.21 +/- 0.03 to 0.13 +/- 0.03 in LNCaP multicellular spheroids, and from 0.13 +/- 0.04 to 0.03 +/- 0.02 in DU 145 multicellular spheroids. This was accompanied by 65% increase in the expression of cytokeratins 8 and 18 in DU 145 spheroids. PSA expression extended 60 micro m within static spheroids and was disrupted in mixed culture. Diminished PSA expression and spatial organization suggests a more aggressive cancer. Higher cytokeratin expression could result from either differentiation towards a luminal phenotype or activation of the Ras pathway during dedifferentiation. Thus, the existing paradigm of differentiation established for normal tissue does not apply for our neoplastic spheroids. Cell dedifferentiation is attributed to improved interstitial transport and synthesis of extracellular matrix.

H Song, KC O'Connor*, DJ Lacks, RM Enmon and SK Jain. Monte Carlo simulation of LNCaP human prostate cancer cell aggregation in liquid-overlay culture. Biotechnol Prog, 19: 1742-9 (2003).

Neoplastic cells self-assemble in liquid-overlay cultures into multicellular spheroids that resemble micrometastases and avascular regions of larger tumors. A Monte Carlo simulation based on Meakin's cluster-cluster aggregation model resolved the physical mechanisms by which LNCaP human prostate cancer cells aggregate in this environment. The best-fit solution suggests that LNCaP cells aggregate with an adhesion probability of 0.5% when they migrate within a radius of influence between cell centers of 180 microm, 10 times the cell diameter. The sweeping radius of influence is indicative of cell tethering and/or chemotaxis and results in an intrinsic rate of self-aggregation that increases from k(11) = 1.5 h(-1) for single cells to k(1010) = 17.5 h(-1) for 10-mers. Similar rates are predicted by Smoluchowski's collision theory (1), suggesting that they are inherent properties of LNCaP liquid-overlay culture. Aggregates form more compact structures in culture than during simulation as measured by the fractal dimension: D(F) = 1.74 +/- 0.04 for 10-mers in culture vs D(F) = 1.25 +/- 0.10 for simulated 10-mers. Additional restructuring would further extend the radius of influence and diminish adhesion. Applications of this work include the production of highly viable spheroids for drug testing and basic oncological research.

H Song, O David, S Clejan, CL Giordano, H Pappas-LeBeau, L Xu and KC O'Connor*. Spatial composition of prostate cancer spheroids in mixed and static cultures. Tissue Eng, 10: 1266-76 (2004). Aggregation of neoplastic cells produces multicellular spheroids resembling micrometastases. The objective of this study was to investigate the effects of mixing culture medium on the spatial composition of spheroids prepared from well (LNCaP) and poorly (DU 145) differentiated human prostate cancer cells. Spheroids were cultured in a mixed suspension within a high-aspect rotating wall vessel and static liquid-overlay plate. Results from this study demonstrate that mixed cultures consistently manifested differences in morphology and composition between DU 145 and LNCaP spheroids. For example, 40 +/- 12% of DU 145 cells were Ki-67 positive 100 microm from the surface within mixed spheroids versus 0% for LNCaP cells; there was no significant difference in this spatial profile for static cultures. The results suggest that poorly differentiated spheroids may be more likely to experience a change in composition from mixing culture medium than well-differentiated spheroids, due to low tissue density. Immunostaining for P-glycoprotein is representative of this trend; average staining intensity increased 50% for DU 145 spheroids on mixing but was unchanged for LNCaP spheroids. The effects of mixing on spheroid composition were attributed to faster interstitial mass transport. Applications include drug development and delivery, as well as basic research on drug action and resistance.

C Vidulescu, S Clejan and KC O'Connor*. Vesicle traffic through intercellular bridges in DU 145 human prostate cancer cells. J Cell Mol Med, 8: 388-96 (2004).

We detected cell-to-cell communication via intercellular bridges in DU 145 human prostate cancer cells by fluorescence microscopy. Since DU 145 cells have deficient gap junctions, intercellular bridges may have a prominent role in the transfer of chemical signals between these cells. In culture, DU 145 cells are contiguous over several cell diameters through filopodial extensions, and directly communicate with adjacent cells across intercellular bridges. These structures range from 100 nm to 5 microm in diameter, and from a few microns to at least 50-100 microm in length. Time-lapse imagery revealed that (1) filopodia rapidly move at a rate of microns per minute to contact neighboring cells and (2) intercellular bridges are conduits for transport of membrane vesicles (1-3 microm in diameter) between adjacent cells. Immunofluorescence detected alpha-tubulin in intercellular bridges and filopodia, indicative of microtubule bundles, greater than a micron in diameter. The functional meaning, interrelationship of these membrane extensions are discussed, along with the significance of these findings for other culture systems such as stem cells. Potential applications of this work include the development of anti-cancer therapies that target intercellular communication and controlling formation of cancer spheroids for drug testing.

H Song, SK Jain, R Enmon, O'Connor*. Restructuring dynamics of DU 145 and LNCaP prostate cancer spheroids. In Vitro Cell Dev Biol - Anim, 40: 262-7 (2004). [HIGHLIGHTED ARTICLE]

Neoplastic cells acquire multidrug resistance as they assemble into multicellular spheroids. Image analysis and Monte Carlo simulation provided insight into the adhesion and motility events during spheroid restructuring in liquid-overlay culture of DU 145 and LNCaP human prostate cancer cells. Irregularly shaped, two-dimensional aggregates restructured through incremental cell movements into three-dimensional spheroids. Of the two cultures examined, restructuring was more pronounced for DU 145 aggregates. Motile DU 145 cells formed spheroids with a minimum cell overlay of 30% for 25-mers as estimated by simulation vs. 5% for adhesive LNCaP cells in aggregates of the same size. Over 72 h, the texture ratio increased from 0.55 ± 0.05 for DU 145 aggregates with projected areas exceeding 2000 ?m2 to a value approaching 0.75 ± 0.02 ($p < 0.05$). For LNCaP aggregates of comparable size, the increase in texture ratio was more modest, less than 15% during the same time period ($p < 0.05$). Combined these data suggest that motility events govern the overall rate of spheroid restructuring. This information has application to the chemosensitization of solid tumors and kinetic modeling of spheroid production.

	KC O'Connor* and M Venczel. Predicting aggregation kinetics of DU145 human prostate cancer cells in liquid-overlay culture. Biotechnol Lett, 27: 1663-8 (2005). This study evaluates the predictive capacity of a novel population-balance model to simulate aggregation kinetics of attachment-dependent cells at the resolution of one-cell increments. Using spheroid assembly of DU 145 human prostate cancer cells as a representative system, the mathematical model proved to be robust in simulating aggregation over a 5-fold range of surface densities from 5 x 10(3) to 2.5 x 10(4) cells/cm2 with a single matrix of rate constants. For cultures at 1 x 10(5) cells/cm2, more than 75% of simulated aggregate concentrations are within the standard deviation of measured concentrations. For the two extreme densities, at least two-thirds of model predictions are within 35% of the mean for experimental data. Error in model predictions is attributed to uncertainty in measurements and intrinsic changes in aggregation. The model has application to the rational design of spheroids in tissue engineering and bioseparation processes in pharmaceutical manufacturing. BL Barrilleaux, M Almagor, H Song, S Japa, O David and KC O'Connor*. Three-dimensional spheroid assembly induces accumulation of E-cadherin in prostate cancer cells. In Vitro Cell Dev Biol – Anim, submitted. Reversible E-cadherin regulation is associated with metastatic progression of carcinomas. E-cadherin is down-regulated during invasion, but can rise again at metastatic sites. To investigate this phenomenon, we examined E-cadherin expression in liquid-overlay culture of prostate cancer cells, which allows cells to form spheroids that mimic the microenvironment seen in vivo during the formation of micrometastases. Immunohistochemical analysis of spheroids showed apparent up-regulation of E-cadherin in DU 145 cells as compared with monolayer culture, while LNCaP cells had strong, uniform staining in both culture types. Additionally, increased accumulation of E-cadherin protein was evident for spheroids of both cell types. ELISA showed that E-cadherin increases from 5.4 +/- 1.0 to 12.2 +/- 0.9 ng/mg total protein between DU 145 monolayers and spheroids ($p < 0.001$), and from 40.0 +/- 7.0 to 54.6 +/- 2.3 ng/mg total protein between LNCaP monolayers and spheroids ($p < 0.01$). RT-PCR, however, revealed no difference between the two culture types at the transcriptional level. Greater accumulation of E-cadherin in spheroids may, to some extent, be due to altered protein trafficking. Given the involvement of adhesion molecules, especially E-cadherin, in metastasis, the data here validate the use of three-dimensional culture to study the mechanism of E-cadherin regulation and potentially the regulation of other adhesion molecules in metastatic progression.

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